

CAC Controller - Installation Guide

Step-by-step installation guide for the CAC Controller on a Raspberry Pi — Pioneer CAC CD Autochanger Web Controller

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1. Overview

Detailed step-by-step guide for installing the CAC Controller on a Raspberry Pi. The controller operates a Pioneer CAC CD autochanger (V3000, V3200, V5000, or V180M) via a serial RS-232C connection.

What is the CAC Controller?

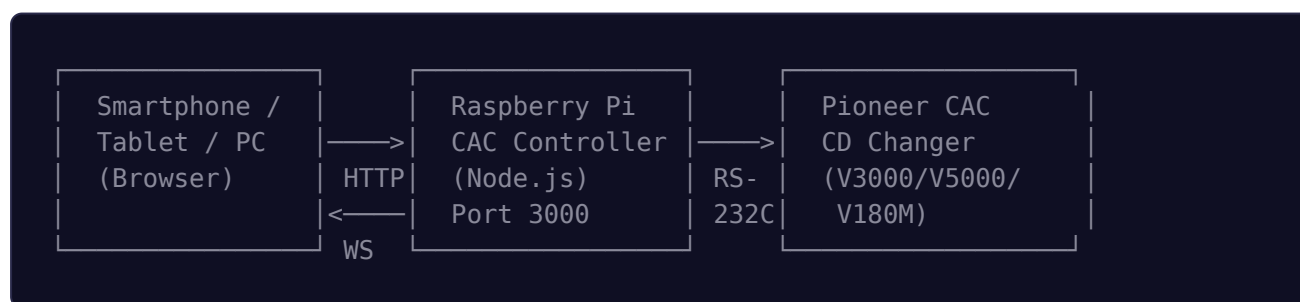
The CAC Controller is a Node.js web application that runs on a Raspberry Pi and controls a Pioneer CAC CD autochanger via the serial interface (RS-232C). It provides:

- — Responsive dark-theme UI for browsers and smartphones
- — Control both players of the changer simultaneously
- — Catalog all CDs with covers, artist info, and track lists
- — Automatic metadata lookup
- — Automatically read and catalog CDs
- — Create custom playback lists
- — Installable as an app on smartphones

Supported Pioneer Models

Model	Slots	Baud Rate	Notes
CAC-V3000	300	9600	Standard reference model
CAC-V3200	300	9600	Same as V3000
CAC-V5000	500	9600	500-disc magazine
CAC-V180M	18	4800	Compact changer, different baud rate

System Architecture



2. Prerequisites

Hardware

- — Recommended: Pi 3B+, Pi 4, Pi 5, or Zero 2 W
 - Pi Zero (first generation) works but is significantly slower during installation
 - Minimum 512 MB RAM
- — Minimum 8 GB, recommended 16 GB or more
- — Appropriate USB power supply for your Pi model
- — USB-A to RS-232C (DB9 or DB25)
 - Chipsets: FTDI FT232R, Prolific PL2303, or CH340 (all Linux-compatible)
 - Will be recognized as `/dev/ttyUSB0`
- — RS-232C cable to the Pioneer changer
 - 15-pin D-Sub connector (Pioneer side)
 - 9-pin D-Sub connector (adapter side) or 25-pin depending on adapter
 - Pinout: TX (Pin 2), RX (Pin 3), GND (Pin 5) — crossed (null modem)
- — WiFi or Ethernet
- — Powered on and operational

Software

- — Lite or Desktop (Debian Bookworm or newer recommended)
- — Enabled (in Raspberry Pi Imager under settings)
- — For installing packages

Network

- The Raspberry Pi must be on the same network as the control device (smartphone/PC)
- Recommended: Static IP address for the Pi (see Section 3)
- Default port: 3000 (configurable)

3. Preparing the Raspberry Pi

3.1 Installing the Operating System

1. Download : <https://www.raspberrypi.com/software/>
2. Flash Raspberry Pi OS Lite (64-bit) to the MicroSD card
3. In the Imager under (gear icon):
 - Set hostname (e.g., `cac-livingroom`)
 - Enable SSH
 - Set username and password (default: `pi`)
 - Configure WiFi (SSID and password)
 - Set locale/timezone

3.2 Connect via SSH

```
ssh pi@<IP_ADDRESS>
```

If the IP is unknown:

```
# From your local machine (right after first boot):  
ping cac-livingroom.local  
# or check your router's DHCP lease list
```

3.3 Update the System

```
sudo apt update && sudo apt upgrade -y
```

3.4 Configure Static IP Address (Recommended)

For Raspberry Pi OS Bookworm (NetworkManager):

```
# Show current connection:  
nmcli connection show  
  
# Set static IP (example for WiFi):  
sudo nmcli connection modify "preconfigured" \  
    ipv4.method manual \  
    ipv4.addresses "192.168.1.100/24" \  
    ipv4.gateway "192.168.1.1" \  
    ipv4.dns "192.168.1.1"
```

```
# Restart connection:
sudo nmcli connection down "preconfigured"
sudo nmcli connection up "preconfigured"
```

Replace the connection name (`preconfigured`) with your actual name (see `nmcli connection show`).

3.5 Disable WiFi Power Saving (Recommended)

Power saving mode can cause delays and connection drops:

```
# Check current status:
iw wlan0 get power_save

# Disable:
sudo iw wlan0 set power_save off
```

Make it permanent:

```
sudo tee /etc/NetworkManager/conf.d/wifi-powersave.conf > /dev/null << 'EOF'
[connection]
wifi.powersave = 2
EOF
sudo systemctl restart NetworkManager
```

4. Installing Node.js

4.1 Install Node.js 20 LTS

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

4.2 Verify Installation

```
node --version
# Expected output: v20.x.x (minimum v18.x)
```

```
npm --version  
# Expected output: 10.x.x
```

4.3 Alternative: Node.js via nvm

If the NodeSource method causes issues:

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.0/install.sh | bash  
source ~/.bashrc  
nvm install 20  
nvm use 20
```

5. Installing Build Tools

The native SQLite extension (`better-sqlite3`) needs to be compiled:

```
sudo apt install -y build-essential python3
```

On a fresh Raspberry Pi OS, these packages are usually already present. This command ensures all required compilers and headers are installed.

6. Setting Up the Serial Port

6.1 Add User to dialout Group

So the CAC Controller can access the serial port:

```
sudo usermod -a -G dialout pi
```

Log out and back in for the group change to take effect:

```
# Either re-login:  
exit
```

```
ssh pi@<IP_ADDRESS>

# Or just update the group:
newgrp dialout
```

Verify group membership:

```
groups
# Should include "dialout"
```

6.2 Connect and Check USB-Serial Adapter

1. Plug the USB-serial adapter into the Raspberry Pi
2. Check if it is recognized:

```
# List USB devices:
lsusb
# Should show the adapter, e.g.:
# Bus 001 Device 003: ID 0403:6001 Future Technology Devices International, Ltd FT232 Serial (UART
# Check serial port:
ls -la /dev/ttyUSB*
# Expected output: /dev/ttyUSB0

# If not found, check kernel log:
dmesg | grep -i tty
```

6.3 Connect the Serial Cable

1. Connect the USB-serial adapter to the serial cable
2. Plug the serial cable into the port on the Pioneer changer
3. Power on the Pioneer changer

6.4 Test Serial Connection (Optional)

Manually verify that communication works:

```
# Configure port (example for V3000 at 9600 baud):
stty -F /dev/ttyUSB0 9600 cs8 -cstopb -parenb

# Query status (sends "Player 1 Status" command):
echo -ne "1PS?X\r" > /dev/ttyUSB0
```

```
# Read response (wait 3 seconds):
timeout 3 cat /dev/ttyUSB0
# Should return a response like "1PS..."
```

7. Installing the CAC Controller

Option A: From GitHub (Recommended)

```
cd /opt
sudo git clone https://github.com/NFDiJee/cac-controller.git
sudo chown -R pi:pi /opt/cac-controller
cd /opt/cac-controller
npm install
```

The `npm install` process takes several minutes on a Raspberry Pi (up to 10–15 minutes on a Pi Zero), as `better-sqlite3` is compiled natively.

Option B: Manual Copy

If Git is not available or preferred:

```
# On your local machine:
scp -r ./cac-controller/* pi@<PI_IP>:/opt/cac-controller/

# On the Raspberry Pi:
cd /opt/cac-controller
npm install
```

Verify Installation

```
ls -la /opt/cac-controller/
# Should contain: server.js, package.json, src/, public/, ...

ls -la /opt/cac-controller/node_modules/
# Should show installed packages
```

8. First Start and Testing

8.1 Start the Server Manually

```
cd /opt/cac-controller  
node server.js
```

Expected output:

```
[Server] Initializing database...  
[Server] CAC Controller running on http://0.0.0.0:3000  
[Server] Model: CAC-V3000  
[Server] Serial: /dev/ttyUSB0 @ 9600 baud
```

8.2 Open in Browser

On your smartphone/PC, open in the browser:

```
http://<PI_IP>:3000
```

Example: `http://192.168.1.100:3000`

You should see the CAC Controller dashboard with:

- Player cards for Player 1 and Player 2
- Transport controls (Play, Stop, Pause, Next, Previous)
- Navigation (Library, Scanner, Playlists, Settings)

8.3 Test First Connection

1. Press `Enter` in the browser
2. The changer should load a CD and start playing
3. The player card should update with status (disc, track, time)

8.4 Stop the Server

Press `Ctrl+C` in the terminal to stop the server.

9. Setting Up the Systemd Service

A systemd service ensures the CAC Controller starts automatically on boot and restarts on crashes.

9.1 Copy Service File

```
sudo cp /opt/cac-controller/cac-controller.service /etc/systemd/system/
```

9.2 Adjust Service File (If Needed)

If your user is not `pi` :

```
sudo nano /etc/systemd/system/cac-controller.service
```

Contents of the service file:

```
[Unit]
Description=CAC Controller - Pioneer CD Autochanger Controller
After=network.target

[Service]
Type=simple
User=pi
Group=pi
WorkingDirectory=/opt/cac-controller
ExecStart=/usr/bin/node server.js
Restart=always
RestartSec=5
Environment=NODE_ENV=production

[Install]
WantedBy=multi-user.target
```

Change `User=` and `Group=` to the correct user.

9.3 Enable and Start the Service

```
# Reload systemd configuration:
sudo systemctl daemon-reload
```

```
# Enable service on boot:
sudo systemctl enable cac-controller

# Start service now:
sudo systemctl start cac-controller
```

9.4 Check Status

```
sudo systemctl status cac-controller
```

Expected output:

```
● cac-controller.service - CAC Controller - Pioneer CD Autochanger Controller
   Loaded: loaded (/etc/systemd/system/cac-controller.service; enabled)
   Active: active (running) since ...
```

9.5 View Logs

```
# Follow current logs live:
sudo journalctl -u cac-controller -f

# Last 50 lines:
sudo journalctl -u cac-controller --no-pager -n 50
```

9.6 Test Autostart

```
sudo reboot
```

After reboot (approx. 30–60 seconds), `http://<PI_IP>:3000` should be accessible again.

10. Configuring Settings

In the browser under [Settings](#) (gear icon), the following settings can be adjusted:

10.1 Serial Port

Setting	Default	Description
Port	/dev/ttyUSB0	Serial port of the USB adapter
Baud Rate	9600	9600 for V3000/V3200/V5000, 4800 for V180M

10.2 Device Settings

Setting	Default	Description
Model	CAC-V3000	Select Pioneer model
Max Discs	300	Number of slots: 300 (V3000/V3200), 500 (V5000), 18 (V180M)

10.3 Network

Setting	Default	Description
Port	3000	HTTP server port
Language	auto	Automatic, German, or English

10.4 Apply Changes

After changes to port or serial settings, restart the service:

```
sudo systemctl restart cac-controller
```

11. Configuring Hub Connection

If a CAC Hub is being used, the node must have an API key.

11.1 Generate API Key

1. In the browser under [CAC Hub](#), open the [API Key](#) section
2. Enter a [Room Name](#) (e.g., "Living Room")

3. Enter a (e.g., "Ground Floor")
4. Click — creates a random 32-character key
5. Click

11.2 Note the API Key

Copy the generated API key — it will be needed later when adding the node in the Hub.

11.3 Test Connection

From the Hub Pi or another machine on the network:

```
curl -H "X-API-Key: YOUR_API_KEY" http://<NODE_IP>:3000/api/node/info
```

Expected response:

```
{
  "name": "Living Room",
  "room": "Ground Floor",
  "model": "CAC-V3000",
  "version": "1.0.0"
}
```

12. Building the CD Database

After installation, the CD library is empty. There are three ways to populate it:

12.1 Automatic CD Scanning

The scanner has the changer load each CD, read the TOC (Table of Contents), and return the CD:

1. In the browser under
2. Enter (e.g., 1)
3. Enter (e.g., 300)
4. Click

Approx. 30–60 seconds per CD (load + read TOC + return). A full scan of 300 slots takes about 3–5 hours.

The scan can be paused at any time and resumed later. Already scanned CDs are preserved.

12.2 MusicBrainz Search

For individual CDs, metadata can be searched automatically:

1. In the browser under
2. Search by or
3. Select a result
4. Apply to a slot

12.3 JSON Import

If a database from a previous installation exists:

1. In the browser under
 2. Select a `cds.json` file
 3. Click
-

13. Installing PWA on Smartphone

The CAC Controller can be installed as a Progressive Web App (PWA) on your smartphone:

iOS (Safari)

1. Open `http://<PI_IP>:3000` in Safari
2. Tap the icon (square with arrow)
3. Select
4. Enter a name and tap

Android (Chrome)

1. Open `http://<PI_IP>:3000` in Chrome
2. Tap the
3. Select
4. Tap

The app then runs in full-screen mode like a native app.

14. Updates

From GitHub Repository

```
cd /opt/cac-controller
git pull
npm install
sudo systemctl restart cac-controller
```

Manual Update

1. Copy new files to the Pi (via SCP or USB)
 2. Run `npm install` (if dependencies changed)
 3. Restart service: `sudo systemctl restart cac-controller`
-

15. Troubleshooting

15.1 Service Does Not Start

```
# Check logs:
sudo journalctl -u cac-controller --no-pager -n 50

# Service status:
```

```
sudo systemctl status cac-controller

# Start manually for detailed error output:
cd /opt/cac-controller && node server.js
```

15.2 Serial Port Not Found

```
# List all USB devices:
lsusb

# List serial ports:
ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null

# Check kernel log:
dmesg | grep -i tty

# Is user in dialout group?
groups
```

Unplug the adapter, wait 5 seconds, plug it back in. If `/dev/ttyUSB1` appears instead of `ttyUSB0` : change the port in settings.

15.3 No Connection to Changer

```
# Manually test serial connection:
stty -F /dev/ttyUSB0 9600 cs8 -cstopb -parenb
echo -ne "1PS?X\r" > /dev/ttyUSB0
timeout 3 cat /dev/ttyUSB0
```

- Wrong cable (not a null modem cable)
- Wrong baud rate (4800 instead of 9600 or vice versa)
- Changer powered off or in standby
- TX/RX swapped

15.4 Port 3000 in Use

```
# Find process on port 3000:
sudo lsof -i :3000
```



```
# If an old process is running, stop the service:
sudo systemctl stop cac-controller
```

Alternatively: choose a different port in settings (e.g., 8080).

15.5 npm install Fails

```
# Check build tools:
gcc --version
python3 --version

# If not installed:
sudo apt install -y build-essential python3

# Delete node_modules and reinstall:
rm -rf node_modules package-lock.json
npm install
```

15.6 WiFi Disconnects

```
# Check power saving mode:
iw wlan0 get power_save

# Disable (see Section 3.5)
sudo iw wlan0 set power_save off
```

15.7 Raspberry Pi Zero: Slow Installation

On a Pi Zero (first generation), `npm install` can take up to 15 minutes. Alternatives:

1. Compile on a more powerful machine and transfer `node_modules/` via SCP
- 2.

```
sudo dphys-swapfile swapoff
sudo nano /etc/dphys-swapfile
# CONF_SWAPSIZE=1024
sudo dphys-swapfile setup
sudo dphys-swapfile swapon
```

16. Uninstallation

Remove Service

```
sudo systemctl stop cac-controller  
sudo systemctl disable cac-controller  
sudo rm /etc/systemd/system/cac-controller.service  
sudo systemctl daemon-reload
```

Delete Files

```
sudo rm -rf /opt/cac-controller
```

Remove Node.js (Optional)

```
sudo apt purge -y nodejs  
sudo rm -rf /usr/lib/node_modules
```